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Direction Finding with UWB and BLE: A Comparative Study

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***Abstract*—Using Angle-of-Arrival based on Phase-Difference-of-Arrival is a recent trend in indoor positioning. Although the basic techniques for direction estimation have been known and used for a long time in different application scenarios (e.g., RADAR), the complexity of Phase-Difference-of-Arrival transceivers discouraged its use for indoor applications. This scenario changed with the recent development of modern chips at competitive prices. Specifically, two ubiquitous technologies in the context of indoor positioning, namely Bluetooth Low Energy (BLE) and Ultra-Wideband (UWB), currently feature Angle-of-Arrival estimation. In this paper we compare these two technologies regarding their potential for indoor positioning based on a fair evaluation of their performance for estimating Angle-of-Arrival in five different realistic setups. The results obtained are compared with existing studies when available. Our results show that Ultra-Wideband is, in general, more accurate and precise, especially when subject to multipath interference. Additionally, we discuss possible approaches to improve the accuracy and precision of Angle-of-Arrival estimation using BLE, such as the impact of increasing the number of antenna elements. By using UWB, we achieved an angular accuracy of up to 5° even under obstructed LOS and multipath, while an overall mean error of nearly 25° was obtained with BLE in an outdoor scenario without obstacles.**

***Index Terms*—AoA, PDoA, BLE 5.1, UWB, Direction Finding**

I. INTRODUCTION

Direction finding is useful in diverse applications such as radio monitoring (searching for sources of interference), security services, military intelligence, communication systems [1], or more generally to estimate the direction of an emitter relative to a specified direction [2]. Currently, the applicability of direction finding with state-of-the-art communication technologies has attracted attention of the industrial and the scientific community. This specially holds for WiFi, Bluetooth Low Energy (BLE) and Ultra-wideband (UWB) technologies. These technologies are currently used in Real-Time Localization Systems (RTLS) [3], mostly based on distance estimations from multiple reference devices (typically called anchors). Using directional information could enhance such systems by reducing the number of anchors, reducing the energy consumption, and increasing accuracy and precision of the overall system. UWB is typically used for high accuracy indoor applications, achieving sub-decimeter accuracy [4] while BLE has typically a localization error in the order of a few meters [3].

Due to their complexity and high prices, Angle-of-Arrival (AoA)/Phase-Difference-of-Arrival (PDoA) transceivers had, so far, limited applicability. In particular, the use of antenna arrays is known to increase costs and complexity [5]. This scenario is changing as there are novel off-the-shelf devices providing PDoA-based angle estimations for BLE and UWB transceivers, such as the Beta PDoA kit from Decawave (UWB) and the SimpleLink™ Angle of Arrival BoosterPack from Texas Instruments™ (BLE). Also, a new standard has been developed for BLE which supports direction finding: Bluetooth Core Specification Version 5.1. We refer to AoA as a method enabling the estimation of the direction of a incoming signal, which can be done, for instance, by using directional antennas. By PDoA, we refer to a specific method for AoA estimation in which the phase difference between two signals is used. The methods will be explained in detail in Section II.

Although the benefits provided by direction finding with these technologies are evident, few evaluations and no comparison among them were found in the literature. In particular, we compare the given technologies in the same real-world setups using commercial off-the-shelf (COTS) modules, which are available for the public in general, but have been only evaluated in ideal environments and/or independently. While evaluations in ideal environments provide an upper bound on the expected performance of the modules, they usually overestimate it, as will be shown in Section V, fostering the need for an evaluation in realistic environments. Additionally, by performing the experiments in the same environments, we guarantee that the influence of multipath interference is the same for both technologies, enabling a fair comparison. In this paper we investigate the potential of AoA-based BLE and UWB systems for indoor positioning with the goal of providing a concise comparison between technologies, which can be used by developers, system designers and researchers in order to decide which technology fits better a certain application scenario. The key contributions of this paper are:

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- an evaluation of COTS modules which are compliant with established standards;
- an extensive evaluation including obstructed Line-of-sight (LOS) and multipath resilience of PDoA-based AoA estimation for UWB and BLE under the same setups and their impact on direction estimation;
- a fair comparison between the two technologies as well as guidelines on how to select the right technology for a specific application scenario.

II. Standards and Overview of Technologies

This section provides useful information about the standards that the devices used in our experiments are compliant with. Thus, it serves as background supporting specific choices we made towards the experiments presented in Section V.

Critical differences between the two technologies used include:

- **Bandwidth:** BLE can transmit in 40 different $2\,\text{MHz}$ channels (only using one at a time) while UWB bandwidth depends on the channel and can vary from $499.2\,\text{MHz}$ to $1354.97\,\text{MHz}$.
- **Frequency bands:** BLE operates in the $2.4\,\text{GHz}$ ISM frequency band ($2.4\,\text{GHz}$-$2.48\,\text{GHz}$) while UWB can operate either in the sub-gigahertz range, centered at $499.2\,\text{MHz}$, or between $3494.4\,\text{MHz}$ and $9984.0\,\text{MHz}$ (center);
- **Power consumption:** the average power consumption of BLE radios when using the minimum ($7.5\,\text{ms}$) connection interval, was estimated in [6] to be $\approx 10\,\text{mW}$, while UWB transmission consumes on average $\approx 280.5\,\text{mW}$ using the present configuration, as will be shown in Section V-A,

A. UWB

In order to estimate distances and angles, UWB transceivers rely on a Channel Impulse Response (CIR), illustrated in Figure 1, which is obtained from the preamble of the UWB packet. The 802.15.4a packet format [7] can be seen in Figure 2. The point in time at which the propagating signal first reaches the transceiver can be obtained accurately because of UWB's wide bandwidth (short pulse duration).

As UWB uses coherent receivers, which are capable of calculating the phase of the incoming signal, the phase of the first path sample can also be calculated by computing the angle of the complex signal. In AoA estimation, PDoA is obtained by subtracting the phases of the signals received at two spaced antennas.

A recently published approach to estimate AoA using UWB transceivers was proposed by Decawave. This company developed a Beta kit for testing purposes comprising a tag and an anchor. The tag consists of a single transceiver and an omnidirectional antenna. The anchor uses two transceivers, each one directly connected to an antenna, i.e., without any switch in between. The two antennas are approximately 2.05 centimeters apart. In order to determine the angle, a tag

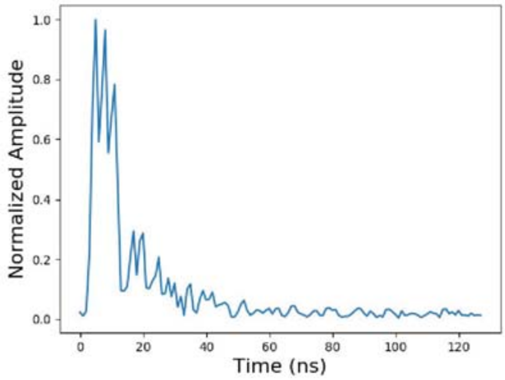

Fig. 1. Typical CIR of an UWB packet. The sample corresponding to the first path can be located within a CIR envelope by a proprietary leading edge detection algorithm with a resolution of 15.6 ps [8] by the chip used in this paper. The total duration of the CIR is 1016 ns $\approx 1\mu s$.

Preamble {16, 64, 1024, 4096} symbols	SFD {8, 64} symbols	PHY header (PHR)	Data

Fig. 2. Standard UWB packet format. The CIR is developed from the symbols transmitted during the preamble (adapted from [7]).

first sends a UWB signal to the anchor. The anchor receives this signal using both of the antennas quasi-simultaneously (there may be a short time delay due to the different path lengths). Each of the (coherent) transceivers is capable of estimating the phase of the signal. Next, the two phases at the *first path sample* are subtracted which results in a so-called *phase difference*. From the phase difference, the processor can calculate the angle-of-arrival by using the well known equation:

$$\theta = \arcsin \frac{\alpha \lambda}{2\pi d}, \tag{1}$$

where α, λ and d denote the phase-difference-of-arrival, the signal wavelength and the distance between the antennas, respectively. A detailed explanation can be found in [9].

B. BLE 5.1

BLE 5.1 core, the most recent standard released by the Bluetooth SIG, features direction finding capabilities. AoA (and similarly Angle-of-Departure) estimation is achieved with BLE by receiving an *unwhitened* signal via an antenna array in which the elements are sequentially switched. The *unwhitened* signal, referred to as Constant Tone Extension (CTE), is a constantly modulated (i.e., without shifts in phase) sequence of 1s which is placed after the BLE payload in the packet. The CTE is optional and has a fixed length varying from $16\,\mu\text{s}$ to $160\,\mu\text{s}$ as described in [10]. The BLE Data Physical Channel PDU Header, illustrated in Figure 3 as in [10], contains a *CP* bit, which if set to one, indicates that this header has a *CTEInfo* appended to it. The *CTEInfo* field indicates the type and length of the CTE. A guard period lasting $4\,\mu\text{s}$ and a reference period lasting $8\,\mu\text{s}$ start the CTE, followed by alternating *switching* and *sampling* slots which can last either $1\,\mu\text{s}$ or $2\,\mu\text{s}$ each. As the name indicates, the receiver should use the *switching*

Header							
LLID (2 bits)	NESN (1 bit)	SN (1 bit)	MD (1 bit)	CP (1 bit)	RFU (2 bits)	Length (8 bits)	CTEInfo (8 bits)

Fig. 3. Header format of the Data Physical Channel PDU (adapted from [10]).

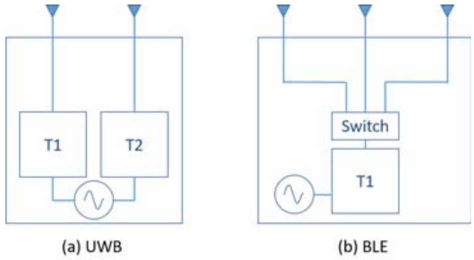

Fig. 4. Transceiver basic architecture for a) UWB and b) BLE. While BLE requires only one transceiver at the module, UWB requires one transceiver per antenna.

slots, to switch its receiving antenna to the next one, and the *sampling* slots to sample the incoming signal.

The standard requires a minimum of 8 reference IQ samples plus at least one (if $2\,\mu$s timeslots are used) additional sample to calculate the AoA estimation, but does not determine how to do so. Only the $2\,\mu$s sample and switch slots are mandatory to support. As a practical reference, the reader is encouraged to read [11]. In Subsection IV-A we detail the approach to estimate the angle from the CTE, as it is not defined by the standard.

Some important differences in the transceivers architectures for both technologies are sketched in Figure 4.

III. Related Work

AoA/PDoA is a well understood and consolidated topic discussed and detailed in several books. The reader may refer to [12] for a detailed overview on the topic. Nonetheless, it is not easy to find comparisons between different standardized technologies performing AoA-based localization nor guidelines covering which technology to choose. In particular, to the best of our knowledge, neither Bluetooth 5.1 nor UWB have been thoroughly evaluated or compared yet regarding AoA performance.

In [13], for instance, the authors compare different algorithms for WiFi only. Also, numerous studies compare this technology without the AoA/PDoA capabilities. In [14], experiments were conducted using Software-defined Radios (SDR) comparing two well-known algorithms for AoA estimation, namely MUSIC and ESPRIT [15], for different numbers of elements in the antennas array. The obtained results are surprising in the sense that increasing the number of elements does not necessarily increase the performance of the algorithm. This result will also be tested once again in this paper, but now with COTS devices. This constitutes an important difference in our paper.

In the next paragraphs we report existing literature related to AoA for 1) Bluetooth 5.1, 2) UWB and 3) a comparison of both technologies.

BLE: In [11], the accuracy of AoA estimation over Bluetooth 5.1 was tested over different frequencies/channels on a Software-Defined Radio (SDR) and the authors showed that frequency diversity improves the accuracy significantly. Using Bluetooth 4.0 and SDRs, it was experimentally demonstrated in [16] that an implemented AoA system comprising two anchors could localize a tag on a 2-D plane with average accuracy of $14\,\mathrm{cm}$. In this paper, we use BLE 5.1 compliant COTS devices, which natively support AoA estimation.

UWB: In the UWB-related literature, one commonly finds claims about the superiority of ultra-wideband-based technologies w.r.t. their narrowband counterparts [9]. We will test within the experiments shown in this paper if this is indeed the case by comparing UWB with BLE, which has a bandwidth considerably narrower than UWB. Two evaluations were found in the literature for UWB-based PDoA systems: [17] and [18]. The first focuses on evaluating a full system using PDoA and not on the angle estimations itself, while the second evaluates a PDoA module focusing at different orientations and elevations for the receiver. None of them evaluates effects of obstructed LOS for PDoA estimation.

UWB vs BLE: A measurement campaign was conducted in [19] with both technologies subject to the same conditions. The BLE localization system used in the evaluation supported PDoA estimation while the one used for UWB supported ranging only. Additionally, only LOS measurements were performed. The authors made the data publicly available, but have not published yet any evaluation nor comparison of the technologies performance. Actually, the goal of the authors is to demonstrate a novel calibration method based on AI with this data. In [20], the authors compared the performance of two COTS localization systems: one using UWB, based on SS-TWR, and the other using BLE (4.0), based on RSSI. Their comparison did not include direction finding capabilities.

In this paper, we are evaluating these technologies regarding their AoA/PDoA performance in different circumstances, such as when the LOS is obstructed and when strongly affected by multipath interference. This paper differs from the previous as it (1) compares AoA capabilities of BLE and UWB under the same conditions, (2) uses COTS devices and not SDR implementation within the experiments and (3) evaluates and compares their performance in a diversity of practical scenarios such as including LOS, obstructed LOS and Multipath.

IV. Experimental Setup and Hardware

Current state-of-the-art off-the-shelf devices were chosen for the evaluation of each technology. They are:

- BLE: SimpleLink™ Bluetooth® Low Energy CC2640R2F wireless MCU LaunchPad™ development kit + SimpleLink™ Angle of Arrival BoosterPack from Texas Instruments™ [21]
- UWB: Beta PDoA kit from Decawave [22]

A. BLE

The only development kit featuring direction finding with BLE 5.1 is the SimpleLink™Angle of Arrival BoosterPack from Texas Instruments™. The same company released an application report [23] and a Node Reference Design [24] reporting AoA estimation accuracy and an evaluation of the antenna for a specific board in an anechoic chamber. The kit was independently evaluated in [25], using an SDK version that was not yet standard compliant. The authors reported promising results in ideal and almost ideal environments, and a significant degradation when testing in realistic environments. Additionally, in [25], two orthogonal antenna arrays were tested consisting of 6 antenna elements in total, which would limit the application of BLE as an anchor in size-constrained devices. We, instead, restricted our experiments to use a single linear array in fairness to UWB. In all our measurements we used 3 antennas sequentially to sample the signals, at a sample rate and switching interval of $1\,\mu$s each, and used a CTE length of $160\,\mu$s. The cost of adding an additional antenna to BLE is relatively low (it does not require an additional transceiver, as is currently the case for UWB). Additionally, we provide a comparison of the benefits and drawbacks of using more or less elements in the same antenna array.

In our experiments, we adopted the same procedure for estimating AoA within BLE as described by Texas Instruments. Only one array of the module was used, containing 3 elements. A switching slot of $1\,\mu$s was used with a sampling frequency of $1\,\mu$s. Thus, for every packet received (with CTE length equals $160\,\mu$s), 74 IQ samples can be acquired, which remain after eliminating the guard period ($4\,\mu$s), the reference period ($8\,\mu$s) and the switching slots ($74 \times 1\,\mu$s) from the $160\,\mu$s CTE. The 74 sampling slots are equally distributed among the three switching antennas meaning that one antenna acquired during 24 slots and two antennas (the first ones) during 25 slots. In order to keep the number of samples evenly distributed over the antenna elements, two slots are discarded, resulting in only 72 samples. We define a *sampling cycle* as the period of time required to acquire 3 samples, in which 1 sample is acquired with each of the 3 antennas. The phase difference between samples of different antennas in the same sampling cycle is calculated and added to a *jitter estimation*. The latter is obtained by calculating the phase difference between two respective samples of the same antenna at consecutive sampling cycles. For instance, consider that the 3 antennas used are tagged as A_1, A_2 and A_3. The Nth phase estimation obtained by A_2, namely $\phi_N(A_2)$, is corrected by the factor $\frac{\phi_{N+1}(A_1)-\phi_N(A_1)}{3}$, which is the phase offset estimated by A_1 during that entire cycle divided by 3. Each antenna pair acquires 23 (offset corrected) estimations during the $160\,\mu$s CTE. Additionally to this correction (which was proposed by the manufacturer), our measurements were fitted by a linear regression model obtained per channel in an anechoic chamber, similar to the approach proposed by the manufacturer. Instead of separating the model per antenna pair, we applied it on the average of the two pairs, as it behaves linearly in certain angular ranges in contrast to one of the pairs. Finally, a moving average window of size 6 was applied to the BLE angle estimations. The size of the window will be discussed in Subsection V-A.

B. UWB

For this technology, Decawave and Ubisense are the only manufacturers (to the best of our knowledge) that developed a module featuring AoA estimation. The module from Ubisense (Dimension4) features 4 antennas while the module from Decawave (dwm1002, shipped as part of the Beta PDoA Kit) features only 2 antennas. A recent study investigated the accuracy of previous localization systems from these companies [26] and concluded that the system from Decawave was superior, despite the system from Ubisense featured AoA already at that time, and the one from Decawave did not. Decawave was superior not only in performance, but also in documentation; Decawave published a diversity of documents about their products including scientific papers, patents and whitepapers/tutorials. No documents regarding Ubisense's method to estimate AoA were found. Therefore, Decawave was chosen for our study.

The receiver module of the Beta PDoA kit is the DWM1002 [22] which comprises two IEEE 802.15.4™-2015 standard-compliant chips, namely, DW1000 [27].

The angle estimations using UWB were calculated as in Equation 1 and corrected by a constant factor calculated by minimizing the error at $0°$. According to the manufacturer this correction is board dependent.

The two technologies achieve direction finding using different approaches. BLE uses a single transceiver and multiple antenna elements, which are connected (one element at a time) via a switch. UWB (we refer to the dwm1002 transceiver already mentioned), uses two transceivers in parallel, each one connected directly to a separate antenna. These different design choices can be easily justified as the AoA capability for BLE is an optional feature (usually BLE is intended for communication), which may or may not be used depending on the application. Also, cost is a reason. For the UWB module, the accuracy was clearly the main priority as receiving the same signal via two spaced antennas avoids errors due to drift in the transmitter's clock. Thus, a higher accuracy is expected from the UWB module. Although the switching approach may be also applicable to UWB, no such implementation was found nor tested. Also, the switching approach may require the tag to send two packets to the PDoA anchor, which increases energy consumption. For BLE, the CTE length can be reduced up to $16\,\mu$s, which is significantly shorter than the minimum packet length ($\approx 84\mu s$). Thus, the addition of a CTE has less impact on energy consumption.

We performed different experiments in order to test the accuracy and precision of the two technologies under LOS, obstructed LOS, strong multipath and both obstructed LOS and strong multipath. In the experiments with obstructed LOS, we introduced different obstacles namely a perfect absorber (a thick layer of absorber foam) and an average-sized human. To test the effect of multipath, we repeated our experiments in two different environments. Our ground-truth measurements for BLE were performed in an anechoic chamber due to a

General Parameters	
Number of antennas	2 (UWB) and 3 (BLE)
Distance between anchor and tag	3m
Obstacle between anchor and tag	None
Number of samples per measurement	1000[1]
Channel	5 (UWB), and all (BLE)

TABLE I
OVERVIEW OF THE PARAMETERS USED BY DEFAULT (IF NOT STATED OTHERWISE) IN THE EXPERIMENTS.

strong non-linearity found in the measurements, which will be discussed in Subsection V-A. For UWB, this was not necessary. Both development kits were evaluated within the angular range $-90°$ to $90°$, where these 2 angles are aligned to the uniform linear array axis and show inferior performance near the fire ends (i.e., near $-90°$ and $90°$, though these angles are omitted from our results). The antenna included in (and soldered to) the Decawave PDoA Beta Kit is directional and covers only its frontal half plane. Therefore, we limited our experiments to this range in order to provide a fair comparison.

Some parameters are common to most of the experiments and will be made explicit otherwise. They are shown in Table I. Two experiments (detailed later) were performed in a seminar room, which is approximately a $6\,\mathrm{m}$ by $8\,\mathrm{m}$ room, with an area of $48.07\,\mathrm{m}^2$ and a height of $3\,\mathrm{m}$. The other two experiments were performed in a long hallway, in the ground floor of the same building which is $3\,\mathrm{m}$ wide and $10\,\mathrm{m}$ long. Channel 5 was selected for UWB as recommended by the manufacturer [2] of the module used, as the separation of the two antennas is less than $\frac{\lambda}{2}$ for this frequency ($fc = 6489.6MHz$) which prevents angular ambiguities. For BLE, all data channels, i.e., channels 0 to 36, were used.

V. EXPERIMENTS

As outlined in Section IV, we evaluated the performance of BLE and UWB in 5 different setups, among which one is exclusively to test the effect of using additional antennas in the BLE technology. A detailed description of each experiment will be given in a dedicated subsection.

A. LOS - BLE vs. UWB

In this experiment we aim to compare both technologies regarding performance of AoA estimation in a typical office environment, sketched in Figure 5. We fixed the position of the tag and anchor and rotated the anchor in a range from $-90°$ to $90°$ in steps of $30°$ [3] and at 3 different distances from the anchor, namely, $2\,\mathrm{m}$, $3\,\mathrm{m}$ and $4\,\mathrm{m}$. The goal of this experiment is to compare the resilience of both technologies to usual multipath interference, and their accuracy and precision in such an environment. The measured angles and variances for BLE and UWB per distance are plotted in Figure 6.

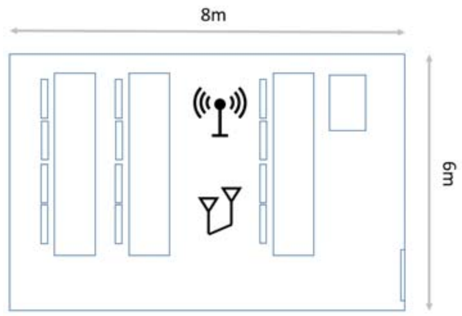

Fig. 5. Layout of the office environment in which the LOS experiment took place.

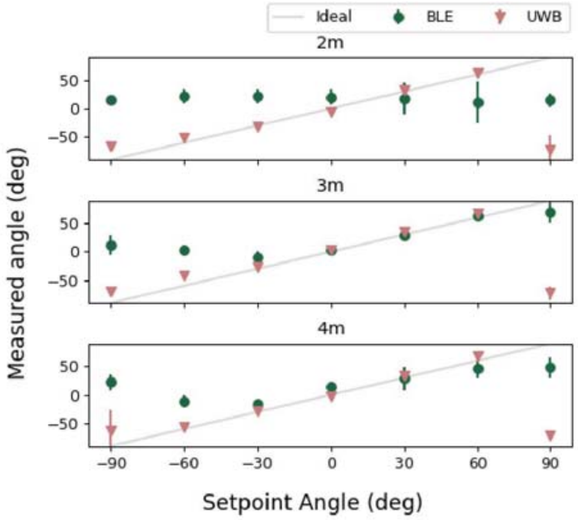

Fig. 6. Angles in steps of $30°$ measured in a seminar room at 3 different distances for the two technologies. A constant offset was added to the all the UWB samples aiming to centralize the measurements at $0°$. BLE data was fitted by a linear regression model obtained by the authors in an anechoic chamber (to minimize multipath reflections and interference) and filtered by a moving average filter (similarly to the out-of-the-box approach provided by the manufacturer). UWB is clearly more accurate and precise. The poor fit obtained with BLE at $2m$ is likely caused by multipath, given the stable spread seen over channels in Figure 7.

The results with UWB are more accurate and precise than the ones obtained using BLE, although at the distance of $3\,\mathrm{m}$ the results with BLE are comparable for small angles, indicating that at this distance/position the BLE measurements were not significantly affected by multipath interference. UWB AoA's measurements showed to be indifferent to the 3 different multipath conditions tested. Put in numbers, the overall average of root mean squared errors (RMSE) for the 3 positions equal $36.8°$ for UWB and $41.9°$ for BLE. Errors can be seen to get larger, as the absolute value of the angle increases and are particularly high at the fire ends ($-90°$ and $90°$), especially for UWB. For this reason, both Texas Instruments and Decawave do not recommend using these angles.

In order to get a better understanding about the impact

[1] 1000 correct angle measurements were acquired independently on the amount of transmissions a technology needed to achieve this.

[2] No benefit was observed when testing indoors channel 7, which has a wider bandwidth.

[3] we exclude from our comparison the values at $-60°$ due to a strong non-linearity found in the BLE antenna near this angle affecting all the channels. It is still plotted for completeness. We verified through experiments in the anechoic chamber that only angles around $-60°$ are affected by this anomaly, whereas angles with smaller absolute value show the correct linear behavior, enabling to use the module for the evaluation. This issue was reported to Texas Instruments™, but, at the moment, it has not been fully solved yet. Our hypothesis is that the antenna module is hard-broken.

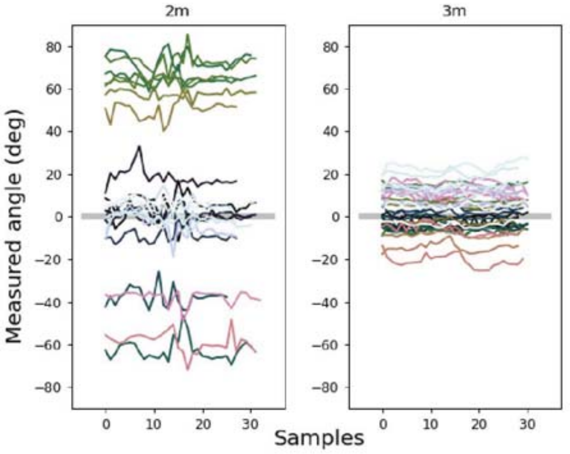

Fig. 7. Dispersion of measurements by channel (in different colors) at 0° at 2 m and 3 m. Channels with variance greater than 25° were filtered out. Measurements are stable by channel (also at $2m$), but spread (in particular, at $2m$), which is likely caused by multipath, as it affects different frequencies in different ways.

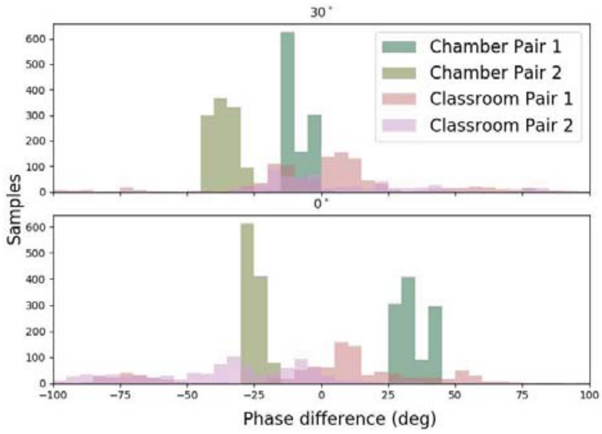

Fig. 8. BLE exemplary phase measurements per antenna pair measured both in the anechoic chamber and indoors for 0° and 30°. When subject to multipath, the data spreads and shifts.

of multipath on BLE angle estimation, we show the raw phase measurements obtained in two different environments, namely classroom and anechoic chamber, in Figure 8. In the classroom, anchor and tag were placed 2 m apart. We can see from Figure 7 that this configuration was likely affected by multipath. The phases were corrected by the jitter estimation only. It can be seen that multipath introduces spread and phase offset, which reduces both accuracy and precision in angle estimations. The histograms also hint at a possible approach of detecting multipath interference, namely by observing the spread in phase. Evaluating this approach is left as future work.

A simple approach can be applied to increase accuracy (not precision), by reducing the standard deviation of BLE measurements, which behave Gaussian-like in most of the experiments: widening the size of the window. However, the minimum BLE connection interval equals 7.5 ms, limiting the time required to acquire a large window, and a BLE packet consumes approximately 75 μJ[6], which results in a higher energy consumption. Considering that a UWB packet with transmitter configured with

- preamble length of 128 symbols;
- SFD of 8 symbols;
- PHR of 16 symbols;
- no data;
- datarate of 6.8 MHz;

(leading to a packet duration of 152 μs) can execute a Two-Way Ranging (TWR) message exchange using our current implementation within 6.25 ms, such an approach would, time-wise, result in a significantly worse performance for BLE. Regarding energy consumption, the transmission of a UWB packet consumes on average $85\,\mathrm{mA} \times 3.3\,\mathrm{V} = 280.5\,\mathrm{mW}$, which multiplied by the duration of 152 μs results in an energy consumption of 42.6 μJ per packet. Therefore, we can conclude that UWB is in any case more efficient, not considering the baseline consumption (only transmission). Considering a typical TWR message exchange, in which a UWB transmitter (tag) is also supposed to receive a response from the receiving side (anchor), the average power estimated [28] for the UWB tag equals 51.9 mW. For the BLE tag it would be estimated [29] to only 4.8 mW with the used configurations, which represents an energy consumption more than 10 times higher for UWB. Another drawback of UWB lies on its anchor, which consumes much more energy ($>$124 μJ per packet).

B. *Obstructed LOS - BLE vs. UWB*

As mentioned in Section III, UWB is thought to be less vulnerable to obstructed LOS because of its wide bandwidth. To test this claim, we inserted (one at a time) a perfect absorber (worst-case scenario, which we also refer to as strongly obstructed LOS) and a human (common obstacle in diverse applications, which we also refer to as lightly obstructed LOS) in the *LOS* setup, i.e., the one from subsection V-A, at an anchor-tag distance of 3 m and analyzed how the angle estimation behaves. At this distance, the multipath interference affected the BLE estimations the least and, therefore, the effect of the obstacle can be observed clearer. The obstacle was positioned always in the middle between anchor and tag, i.e., at a distance of 1.5 m from both. The results of experiments at 3 different angles with the perfect absorber are shown in Figure 9 while the results with the human as obstacle are shown in Figure 10.

From Figure 9, we can see that the perfect absorber did not affect the angle estimations significantly for both technologies and that the error is slightly lower for BLE than for UWB, which is in agreement with Figure 6. The mean RMSE with this obstacle for UWB and BLE equals 9.8° and 8.8°, respectively. Surprisingly, the errors were higher (18.9°) for BLE when the obstacle was a human body, corroborating that the perfect absorber used is not ideal at the frequencies used. For UWB, the mean RMSE in this scenario equals 8.7°, corroborating the idea that UWB angle estimations are also resilient to obstructed LOS.

In order to test if the higher error affecting BLE was caused by interference with an existing WiFi network, we discarded the packets transmitted over all possible overlapping channels, which did not have a significant impact.

The overall mean RMSE of BLE obtained from Experiment V-A (without any obstacle) at $3\,\mathrm{m}$ equals $5.67°$ and was almost evenly distributed over the 3 angles measured ($-30°$ to $30°$). Compared to the results obtained with the two different obstacles in the same position, it can be seen that the overall error always increases, indicating that indeed the performance of this technology gets worse under obstructed LOS. The mean RMSE obtained with UWB for the same conditions without obstacles equals $8.0°$, which is close to the mean RMSE obtained with the human obstacle ($8.7°$) and also from the one obtained with the perfect absorber ($9.8°$). Therefore, the errors for UWB also increase, but relatively less than for BLE.

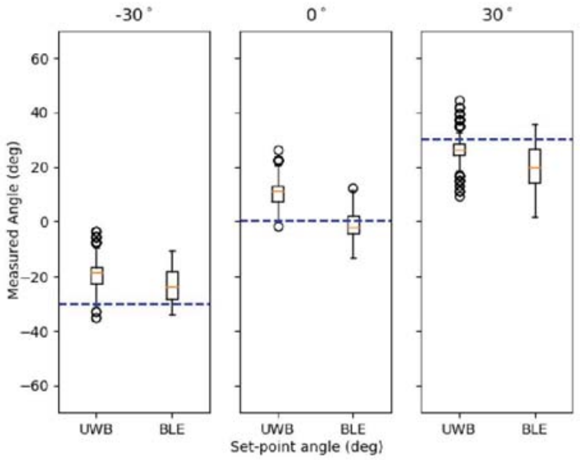

Fig. 9. Distribution of measurement per technology when a perfect absorber obstructs the direct LOS between anchor and tag by being placed exactly in between them. The experiment was performed in a seminar room and the distance between anchor and tag equals $3\,\mathrm{m}$. BLE measurements were filtered by a moving average filter of size 6 (please, refer to Subsection V-A).

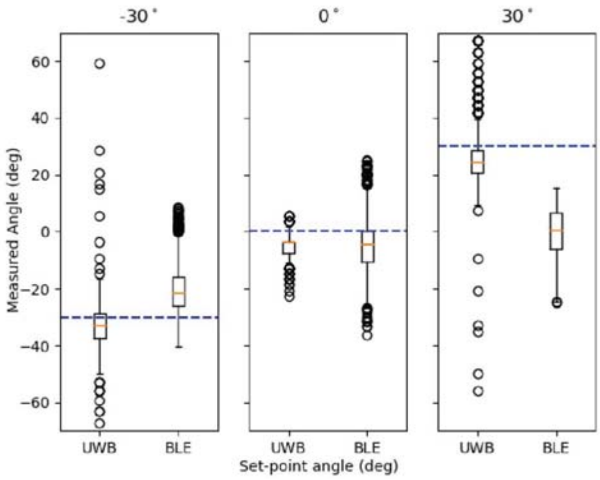

Fig. 10. Distribution of measurement per technology when a human obstructs the direct LOS between anchor and tag by being placed exactly in between them. The distance between anchor and tag equals $3\,\mathrm{m}$. BLE measurements were filtered by a moving average filter of size 6 (please, refer to Subsection V-A).

C. Strong Multipath - BLE vs. UWB

The environment in which the experiment takes place is modified in this experiment. We move to a narrow ($3\,\mathrm{m}$ wide) hallway to increase multipath interference. The distance between tag and antenna is increased to $10\,\mathrm{m}$ so that the time interval between the arrival of the first path and the multipath components is reduced. Specifically, at an anchor-tag distance of $10\,\mathrm{m}$, the time interval between the direct path component and the first multipath component equals $1.467\,\mathrm{ns}$ and is lower than the coherence time for UWB in channel 5 ($\approx 2ns$). Thus, UWB is also subject to multipath interference. We test again for 3 different angles between anchor and tag: $-30°$, $0°$ and $30°$.

According to Decawave [9], the performance of UWB is superior due to its wide bandwidth, which in the time domain manifests as very short pulses. Thus, the multipath components can be separated by the receiver. The bandwidth of BLE, on the other hand, is much narrower. In this experiment, we compare their performance under strong multipath. The results of this experiment can be seen in Figure 11. The bars in orange correspond to BLE measurements performed on all the 37 data channels and show a better RMSE even than UWB, except for the measurements at $30°$, which have a higher error. Since the environment used is densely covered by WiFi access points, which share spectrum with BLE, we test if interference with this technology is the cause for this discrepancy. An additional bar representing the same BLE measurements but discarding the packets obtained on channels overlapping with WiFi was plotted in green. By observing the BLE measurements at $30°$ with and without channels overlapping with WiFi, it gets clear that there is a big improvement, corroborating the hypothesis. Although the measurements excluding WiFi overlapping channels got more uniform, the RMSE got worse at the other angles. This can be justified by the results from [11], which showed that frequency diversity improves the PDoA estimation. Therefore, channel blacklisting can also impose a significant drawback to PDoA estimation, if not carefully implemented. A dynamic channel selection mechanism may be able to address both issues.

In general, it seems that the performance of BLE is comparable to UWB in this scenario. However, the errors for UWB were always negative, i.e., biased in the same direction. By recalibrating the module, it was possible to reduce the errors to less than $5°$ for all the angles tested. The recalibration was made by minimizing the error at $0°$. This hints at two possible approaches to be investigated that may improve performance in UWB: periodical recalibration and calibrating the module offline for different environments.

D. Multipath and obstructed LOS - BLE vs. UWB

This experiment combines the two previous ones, namely obstructed LOS and multipath interference. We place a perfect absorber in between tag and anchor, and perform the experiment in the same environment as in the *strong multipath experiment* (Subsection V-C) but separating the modules by

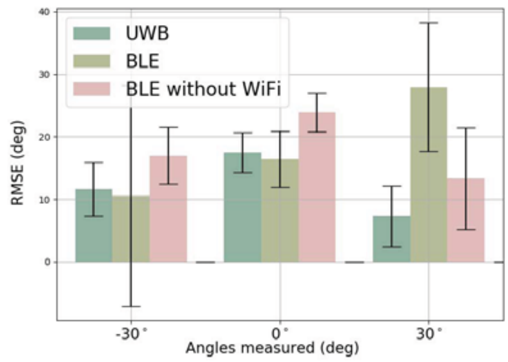

Fig. 11. Mean error and standard deviation of the AoA for the two technologies and the 3 angles observed in a long hallway. The distance between TX and RX equals $10\,\mathrm{m}$ and there is no obstacle in between them. The bar in green corresponds to the same BLE measurements but discarding the packets obtained at channels overlapping with WiFi.

only $3\,\mathrm{m}$. At this distance, the time interval between the direct path component and the first multipath component equals $4.14\,\mathrm{ns}$ and is higher than the coherence time for UWB in channel 5 ($\approx 2ns$). This allows to check whether the performance of UWB was in fact compromised by multipath interference, as we already know from Experiment V-B that UWB is quite resilient to the tested obstacles. Additionally, we evaluate the effect of adding obstacles in a scenario in which BLE is impacted by multipath interference to a known extent.

The mean error regarding the orientation of the anchor towards the tag is shown in Figure 12.

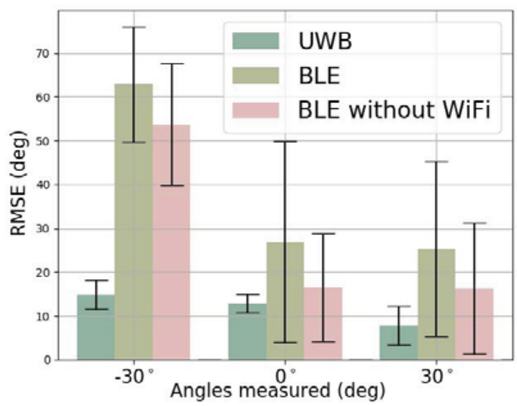

Fig. 12. Barchart showing the errors and standard deviations of the two technologies when positioned in a narrow hallway 3m apart (distance between TX and RX) with a perfect absorber in between.

The same observation as in Subsection V-C regarding the high error for UWB is also valid here: by subtracting a fixed offset, the errors for UWB get even lower. The high error for BLE at $-30°$ could not be significantly improved by blacklisting the channels overlapping with WiFi, but discarding these channels improved the error in the three cases, indicating that interference with WiFi may have been an issue, which in this scenario plays a more important role than channel diversity.

E. BLE - 2 vs. 3 antennas

The performance of AoA estimation is expected to increase with the number of antennas used for sampling due to the spatial diversity introduced. The used development kit provides means to test this improvement in a single linear array by switching among 3 different antenna elements. To our knowledge, no evaluation of the impact of the number of antennas on COTS devices was made public for BLE 5.1 yet or was controversial [14], as the improvement in AoA estimations by increasing the number of antennas was not clear. In this experiment, we intend to compare the deterioration in AoA accuracy and precision by using 2 antennas instead of 3. The cost of adding an additional antenna in the BLE module results mainly from space (in the development kit used it requires additional $35\,\mathrm{mm}$) and possibly a more expensive antenna switch, supporting the desired number of antennas. We analyzed the data that had already been collected in the *LOS experiment* at different angles (from $0°$ to $60°$ in steps of $30°$) when the tag and anchor are positioned $3\,\mathrm{m}$ apart, as we already know that at this position the effect of multipath was minimized.

The measurement for each angle was performed using the 3 antenna elements, and we evaluated in post-processing the performance of using only two elements by discarding the samples obtained from the third element. For fairness, we also eliminated one third of the samples when analyzing the results with 3 antennas, so that the hypothesis of improvement due to using a higher number of samples can be discarded.

We show in Figure 13 the mean error and its standard deviation for the angles mentioned.

It can be seen that by using the two antenna pairs the estimations were improved. Nonetheless, for the angular range shown, the antenna pair plotted alone had a slightly non-linear behaviour w.r.t. the other antenna pair, which boosts the error. Still, the other antenna pair, which behaves linearly in the given range, could not deliver better results than the two pairs combined for any of the angles. By repeating the previous experiments analysis (from V-B to V-D) with only the linear antenna pair, the results in the scenarios affected by multipath got worse. Please, recall that these experiments were performed in the range from $-30°$ to $30°$ in steps of $30°$, in which both pairs behave quite linearly. Therefore, it can be concluded that the inclusion of spatial diversity improves accuracy, especially in scenarios affected by multipath interference. Antennas enabling a more linear mapping from phase difference to angle may improve the results and are currently a focus of research.

VI. Conclusions

Our results showed that UWB performs, in general, direction estimations with a higher accuracy and precision than BLE. In particular, UWB has shown to be more resilient to obstructed LOS and complex environments introducing multipath components. It has been shown, though, that BLE can achieve accurate and stable measurements if not severely affected by multipath interference. In order to improve the performance of this technology in realistic indoor environments, further research is required on the applicability of other known algorithms capable to mitigate multipath interference (e.g., ESPRIT and MUSIC) and possibly on new algorithms.

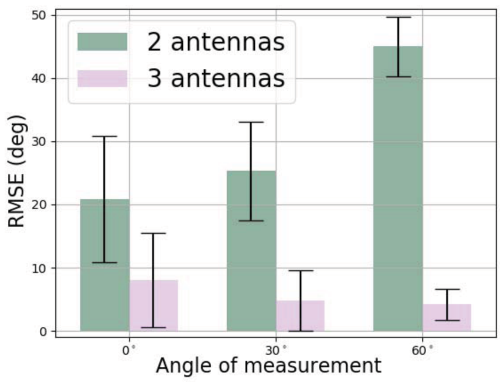

Fig. 13. RMSE and standard deviation when estimating the given angles using BLE with either 2 or 3 antennas in the LOS setup.

These algorithms may probably be executed on constrained platforms regarding size (and therefore antenna number) and processing power, constituting additional challenges in the field. Other factors that can negatively affect BLE estimations are interference with existing WiFi networks, which can be mitigated, for instance, by channel blacklisting, and obstacles within the direct path towards the anchor. However, channel blacklisting reduces the frequency diversity, and, consequently, can reduce the accuracy and precision of angle estimations, in agreement with previous research, and must be carefully implemented. We consider channel blacklisting approaches for mitigating WiFi interference on AoA estimations a relevant topic for future research.

As expected, the benefit of using multiple antenna pairs was observed in diverse experiments with BLE, but further research is still required in order to make this technology suitable for high accuracy applications in more challenging environments, such as decimeter-level indoor localization. This statement also holds for UWB, as errors in the order of $5°$ can, for instance, translate to a position error larger than $30\,\mathrm{cm}$ at a distance of only $4\,\mathrm{m}$ from the anchor. UWB can be suitable for scenarios demanding high accuracy in which the tags are close to the anchor. For example, maintenance in a given machine (where the anchor is attached to) and the goal is to localize the maintenance team (wearing the tags). Special care must be taken with the fire ends of the antennas, which may demand more complex geometries. In the absence of multipath interference and under the same constraints, BLE may be a better option due to its ubiquitousness, as this technology is present in most of the current smartphones and UWB is still not. This constitutes an important advantage of BLE-based localization.

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